

Final Exam Schedule Optimizer

User Manual

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Preparing the Dataset

The web application tool requires two types of datasets for every semester. One is the Course Information Data, which consists of the course identification number, the course title, the instructor's name, and meeting times, among other details. The second is the Student Enrollment Data, which maps every student to their enrolled courses. Both datasets must be prepared in a separate CSV file.

Course Information Data

The table below summarizes the column names and the description for the course information data CSV file. Each course section must have at least one entry in the CSV file.

Default Column Name	Description	Format
Course Reference Number	Reference number that identifies a course.	Sequence of numbers (e.g., 12345)
Course Identification	A course ID that identifies the course more easily than the Course Reference Number.	A combination of a subject code and a course number with no space in between. (e.g., MATH101)
Subject	Subject (e.g. Department) Code of the course.	A sequence of alphabetical letters. (e.g., MATH)
Course Number	Course number of the subject.	A sequence of numbers. (e.g., 101)
Section	Course section number	A sequence of numbers. (e.g., 01)
Primary Instructor Name	Name of the primary instructor of the course.	A sequence of ASCII characters (English alphabet, numbers, and some special characters)
Schedule Description	Describes the schedule type. Note: It is used when the Exam Scheduler suggests a course grouping, which users can later update.	It can be: • "Lecture", • "Lecture/Lab", • "Common Hour", or • "Independent Study"

	This column does not need to be precise for the Exam Scheduler to run.	
Gradable Ind	Describes if a course is gradable or not. If a course is assigned as "Not Gradable," the Exam Scheduler automatically removes the course from final exam scheduling.	It can be: • "Yes", or • "Not Gradable"
Course Title	Title of the course	A sequence of ASCII characters (English alphabet, numbers, and some special characters)
Meeting Days	Days of the week that the courses meet. If this field is empty, the Exam Scheduler automatically removes the course from final exam scheduling.	A combination of the day codes. Day codes: • "M": Monday • "T": Tuesday • "W": Wednesday • "R": Thursday • "F": Friday If a course meets every Monday, Wednesday, and Friday, this field should be "MWF". If a course meets every Tuesday and Friday, this field should be "TF".
Section Begin Time	Begin time of the course.	The format is "HH:MM (AM/PM)". For example, if a course begins at 11 am, this field should be "11:00 AM"

Having irregular meeting times

Some courses may have several meeting times. For example, a chemistry course may have a lecture on every Monday, Wednesday, and Friday, and a lab on Tuesday. A

Spanish course may have a lecture on Tuesday and Thursday, and a recitation on Monday and Wednesday. The users can enter the same course multiple times with different meeting times to describe complex meeting times in the Course Information Data CSV file.

Below table shows the example Spanish course, which is offered every Tuesday and Thursday at 11 AM for a lecture, and Monday and Wednesday at 2 PM for a recitation. When describing a complex meeting times with multiple entries, all fields, except for the Meeting Days and Section Begin Time must remain the same in both entries.

Course Identification	...	Course Title	Meeting Days	Section Begin Time
SPAN101	...	Introduction to Spanish	TR	11:00 AM
SPAN101	...	Introduction to Spanish	MW	2:00 PM

Student Enrollment Data

The table below summarizes the column names and the description for the Student Enrollment Data CSV file. Each student must have one entry in this table.

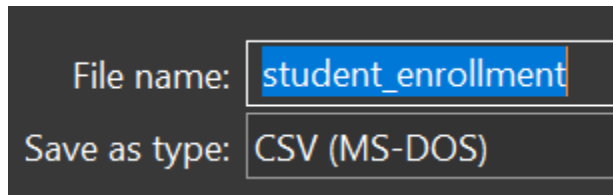
Default Column Name	Description	Format
Randomized ID	Randomized Student ID	Sequence of numbers (e.g., 2839182238910)
CRN 1	Course Reference Number of the first course student is taking. This should map to a corresponding entry in the Course Information Data CSV file.	Sequence of numbers (e.g., 12345) or blank if student is not enrolled in a course.
CRN 2	Course Reference Number of the second course student is taking.	Sequence of numbers (e.g., 12345) or blank if student is not enrolled in a course.
...	...	...
CRN 15	Course Reference Number of the 15th course student is taking.	Sequence of numbers (e.g., 12345) or blank if student is not enrolled in a

		course.
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Note that the Exam Scheduler assumes a student will not take more than 15 courses a semester. If a student takes 4 courses a semester, fields “CRN 1” to “CRN 4” must have a corresponding Course Reference Numbers of the courses, while “CRN 5” to “CRN 15” should remain blank.

Editing the CSV File

You may edit the CSV file in Microsoft Excel, which is recommended if you are manually editing the file. When doing so, make sure to save the data as a CSV File when you save file.



Sample Dataset

This repository contains a sample Course Information Data and Student Enrollment Data CSV files to understand the structure of the dataset. These are under `sample_dataset/` directory. This dataset is too small to test the Exam Scheduler, but it explains the main idea of the dataset structure.

Opening the Application

The application should be opened using the instructions in the README. Briefly, this includes:

1. Installing the dependencies
2. Navigating to the proper directory
3. Running python [manage.py](#) runserver
4. Access the webpage <http://127.0.0.1:8000> in your browser.

Once this is finished, the application should be accessible in the browser for use.

Creating a Semester

Each actual semester in the year will require a new semester to be created in the application. This is to ensure that the application is creating a schedule for the proper student registration and course list. To create a new semester, click on “STEP 1: Create Semester” found on the top bar of the page. This page lists all semesters that you currently have. Click “Load more semester data” to create new semester data from the page shown below.

This is the first page of the exam scheduling tool, where you upload the "exam scheduling" spreadsheet created by the registrar's office with two tabs: one listing course enrollments by student and one listing information about each course.

1. Upload the exam scheduling spreadsheet by selecting it in the "Data File" box and then clicking "Upload"
2. Fill in the Semester's name, which must exactly match the semester's name in the "Academic Period Description" column. E.g. this will look like "Spring Term 2022-2023"
3. Select the exam start date (the first day of exams, after reading day) and end date (the last day of exams)
4. Enter any "Special Courses" in the Special Courses box. Courses should be in all caps with no space separating the department and number, and then separated by a comma.

Example: MATH201, ECON101, SIGN101-02, POLS140-02/03

These are the courses that will be individually scheduled in an exam slot. If all sections of a course will be assigned a common slot, just enter the course name (e.g. MATH201). If just one section of a course will have its own slot, enter the individual section with a hyphen (e.g. SIGN101-02 to denote section 02 of SIGN101). If some - but not all - sections will have a common exam slot, separate those sections with a slash (e.g. POLS140-02/03 for just sections 03 and 04 of POLS140). For reference, you can quickly copy and paste lists from previous semesters and then edit them.

5. Once all information is entered, click the "Go to settings" button below.

Load Final Exam Schedule Data

[Go to settings](#)

Warning: Current data will be deleted if you load a new data

Semester's name: Must match with the Excel's "Academic Period Description" field

1.

Exam start date:

2.

Exam end date:

3.

Exam start times:

4.

Special Courses:

Enter course groups separated by comma. **Examples: MATH201, ECON101, SIGN101-02, POLS140-02/03**

5.

Courses without final exams:

Enter CRNs separated by comma. **Examples: 54920, 52687, ...**

6.

7. Student Data (must be .csv file): No file chosen

8. Course Data (must be .csv file): No file chosen

9. Spring 2023 Special courses

RIQI202, CEEG242, CHEG310, CSC1203, CSC1206, CSC1308, CSC1476, ECON101, ECON127, ECON202, ECON203, ECON204, ECON237, ECON241, ENGR101, ENGR214, ENGR222, GEO1203, GEO1204

1. **Semester name:** something that you will remember what this semester is
 - a. The name used for the semester must not be the name of an existing semester
2. **Exam start date:** the first day that exams will take place
3. **Exam end date:** the last day that exams will take place
 - a. The number of days for exams must be between 4 and 10 days

- b. The start date must be before the end date
- 4. **Exam start times:** the specific times that exams start each day
 - a. There must be 1 to 5 exam times provided and separated by a comma
 - b. Each exam time must be in the format HH:MM AM/PM. Ex: 11:00 AM, 3:30 PM
 - c. Exam times must be between 6:00 AM and 9:00 PM
 - d. Exam times must be at least 3 hours apart from one another
- 5. **Special courses:** courses that should NOT be grouped into the larger course groups and should be scheduled separately
- 6. **(Optional) Courses without exams:** any courses that don't require a scheduled exam. The application will try its best to guess which courses don't have exams.
- 7. **Student Registration CSV:** A CSV file containing all students and their registered courses. This file will be generated automatically and will be located on the computer that is running this application.
 - a. To find this file, click "Choose file". A file window will open. Click "This PC", then "local disk", "Net Files", and finally "Final_Exam_Data". The file will be in this folder
- 8. **Course Information CSV:** A CSV file containing what courses are offered and their instructor. This file will be located in the same place that the student registration is found
 - a. To find this file, click "Choose file". A file window will open. Click "This PC", then "local disk", "Net Files", and finally "Final_Exam_Data". The file will be in this folder

Viewing Semesters

During semester creation, the application guesses how to group courses together. For the best outcomes, these should be manually verified. On the screen listing semesters (click “STEP1” at the top to get there), click “Edit Course Groups” to see the page shown. This page allows you to see information about a specific semester and change its course groups.

Final Exam Scheduler STEP 1: Create a Semester STEP 2: Optimize a schedule STEP 3: Analyze schedules

This page allows you to enter which section don't have an exam and to double-check/edit which course group each section is assigned to. By default, each course is assigned:

- NO_EXAM if it does not have an assigned meeting time in the exam scheduling spreadsheet you uploaded
- Its course/section name, if it is a special course that should be assigned to its own slot (e.g. MATH201, SIGN101-02, etc).
- Otherwise, if it is not assigned to NO_EXAM or its course/section name, our best guess of which meeting time its exam slot should be associated with. This is flagged as “Clear” in the first column if there is a unique meeting time, and “Ambiguous” if there are multiple times.

On this page, you should update and verify the Course Group of each section:

1. Update the course group to be NO_EXAM for any class that will not offer an exam.
2. Check the course group of all other classes and verify that it is correct or update it to the correct slot. Slots should be entered in all caps with 0 spaces/no colon and in military time (e.g. MWF1300 for classes that should be grouped in the MWF 1:00pm class block for exam slots). Special courses should remain assigned to their course group (e.g. MATH201) rather than being assigned a time. As you type, the system will help autofill course groups (e.g. as you type MWF1, you will see all options beginning with MWF1).
3. Later, if you want to quickly search for and change an individual course, you can use the search bar at the top of the “Course Groupings” box. To go back to seeing all courses, clear the search box (so that it is blank) and hit “Search” again.

Key information Switch semester 1.

Academic Period	Spring Term 2022-2023
Semester Name	Spring Term 2022-2023
Exam Start Date	05/03/2023 (Wed)
Exam End Date	05/10/2023 (Wed)
Exam Times	8:00 AM, 11:45 AM, 3:30 PM, 7:30 PM

If you want to change any of the settings above, add or remove courses, or edit students' enrollment, please re-load the data.

Course Groupings

Export the course grouping File name json

Import the course grouping Choose File No file chosen

Show overlaps between courses

Search via course number (e.g. MATH201)

Show only ambiguous courses: Search

1160 entries found.

Ambiguous	CRN	Course ID/Section	Course Title	Instructor	Meeting times	Course Group	
Clear	50008	CHEM231 01	Analytical Chemistry	Collins, Douglas	TR 10:00 AM	TR1000	Confirm
Clear	50014	CHEM322 01	Inorganic Chemistry II	Smith, Brian	MW 03:00 PM	MW1500	Confirm
Clear	50019	CHEM342 01	Physical Chemistry II	Lam, Yan Choi	TR 10:00 AM	TR1000	Confirm
Clear	50027	MATH192 01	Topics in Calculus	Cyr, Van	MWF 08:00 AM	R1900	Confirm

1. **Switch semester:** return to the previous page to pick a different semester
2. **Semester name:** see the name of the semester being viewed
3. **Exam start date:** the first day of exams for the semester
4. **Exam end date:** the last day of exams for the semester
5. **Exam times:** times for each day of exams when exams start
6. **Export course Grouping:** Allows you to save course groupings and apply them to other semesters. Enter a name, then hit the button to save the current groupings

7. **Import course grouping:** click “choose file” and then select the file that was created using the “export course grouping” button. This will attempt to match this semester to the saved groupings, saving you time from redoing it.
8. **Show overlap between courses:** opens a table with courses listed across and down. Each cell shows the number of students in both of the course groups. For example, if Column “MATH201” and “PHYS 211” had a value of 32, then 32 students are in both courses.
9. **Search bar:** allows you to search for the courses to see their groupings. Select a search method from the drop-down, enter the prompt into the box, then hit search
10. **List of courses:** shows each course that matches the current search (by default, all courses)

Suggested way to update Course Groupings

1. Check “Show only ambiguous courses” and hit search so you only see the potential problems
2. Check each of the courses that appear. If the group is correct, hit confirm next to the course
3. If the group is not correct, click into the text box and delete what is there. Start typing in the correct group until it appears in the drop-down menu. Select the correct course and then hit confirm.
4. (Optional) When all course groups are labeled “clear”, you can download the current course groups to be used if you ever recreate this semester. This saves you from having to check all course groups again.
 - a. To do so, there is a textbox at the top of the “Course Groupings” section that says “File name”.json. Type in a name such that you can remember what course groups this is for, then hit export course groups.
 - b. To load these back in later, use the “choose file” field right below to select the file that was downloaded. Then hit “Import the course grouping” to use the same course groupings that were downloaded.

Creating a Schedule

Creating a new schedule is where the application attempts to place courses in an optimal way, using the information provided both by the selected semester and during schedule creation.

Final Exam Scheduler STEP 1: Create a Semester STEP 2: Optimize a schedule STEP 3: Analyze schedules

1

This page prepares potential schedules for you and allows you to "hardcode" any assignments in (e.g. assigning all RESC098 and FOUN098 sections to Friday at 3:30pm).

1. Use the Initial Constraints option to assign specific exam slots to course groups (e.g. assigning all RESC098 and FOUN098 sections to Friday at 3:30pm). Fill in the Course Group (e.g. "FOUN098") and then select the time slot you would like to assign it.
2. If you want to specify more courses, click the "Add more constraints" button
3. **Note:** You cannot have any empty constraints. If there are lingering constraints that are left blank (i.e. which you don't want to use), click "Remove Constraint"
4. Click "Optimize". Then wait while the system creates and analyzes the schedule.

Initial Settings

1. Schedule Name

2. Semester

Spring Term 2023-2024

3. Optimization Preference

spring 2024 survey results

4. Manage Preference Profiles

5. List of Course Groups

Show Groups

6. Constraints

Course Group

Assigned time slot

7. Add more constraints

7. Course groups that should not be placed on the last day of the exam

8. Course groups that should not be placed on the last 2 days of the exam

9. Course groups that should not be placed at night

10. Course groups that should not be placed on Friday and Monday

Optimize

1. **Schedule name:** a name to help you remember this schedule
2. **Semester:** The semester you want this schedule to optimize for. This semester must have been previously created
3. **Optimization preference:** While optimizing, each "problem" that a schedule can have (i.e. overlaps) has a penalty associated with it. By selecting a preference profile, you can tell the application the relative penalties that these different problems should incur. If you find that you want to reduce back-to-backs, pick a preference profile that has a **higher value** for the back to back entry.
4. **Manage Preference Profile:** Takes you to a page where you can view, create, and delete Preference Profiles to be used during optimization.

5. **Show Groups:** Button to show/hide the list of groups that can be entered into field 6 through 10
6. **Constraints:** If a course needs to be put at a specific time, this is where to set it. Click “Add more constraints”, then pick a course group from the list (typing helps narrow the selection), then pick what time that course should be assigned to. If more courses need to be fixed like this, click add constraint again.
7. - 10. **General restrictions:** These are the fields at the bottom of the schedule creation. They let you block courses from being scheduled at certain times. For example, entering MATH211 into the field “Course groups that should not be placed on the last day of the exam” will prevent MATH 211 from being placed on the last day.
8. **Optimize:** Once all the information is entered. Click this button to begin optimization

The application will then return you to the dashboard to see other schedules you have created. Optimization takes a long time, around 30 or 40 minutes. To see updates as to what stage of optimization the application is in, **refresh the page** and check the “Status” column for the schedule you created.

Some notes:

- If the status is reported to be “Infeasible”, that means there were conflicting constraints. An example of this would be setting MATH211 to be scheduled on Friday at 8:00 AM, but then also entering MATH211 into the field “Course groups that should not be placed on Friday and Monday”. Delete this infeasible schedule and create a new schedule without the conflicting constraints.
- If the status is reported as “Error,” the application encountered a problem during optimization. Try closing the application, reopening it, and then recreating the schedule. If problems persist, try changing and constraints that were added during schedule creation.
- After around 30 or 40 minutes, refresh the page again. By now the schedule should have a view button to let you analyze the schedule. If the status is not “Error” or “Infeasible,” then wait and refresh the page later for the view button to appear.
- Preference Profiles are viewed, created, and deleted by clicking on “Manage Preference Profiles” in the schedule creation page.

Analyze a Schedule

Once the page has been refreshed and the schedule has a view button next to it, click this button to get a detailed view of the schedule. There are a couple of important sections to this page:

Schedule Details

1.

Return to Dashboard

Create Spreadsheet

Download

2.

3.

5.

6.

7.

8.

9.

10.

11.

Schedule Title

grasp testing part a billion

Semester

Spring Term 2023-2024

Version Name

New Version ☒

Select version

New Version

Delete

Profile Preference

Show Profile Preference

Constraints

Show Constraints

Final exam date and time	05/01 (Wed)	05/02 (Thu)	05/03 (Fri)	05/06 (Mon)	05/07 (Tue)	05/08 (Wed)
8:00 AM	MWF0900 MW0830 WF0830	MWF1500 MW1500	MWF1300 MW1330 M1300 F1300 MWF1330	TR0830 MW1300 CEE6242 BIOL202 T0830	MWF1000 MW0900 MW1000	MWF1100
11:45 AM	ECON204 PHIL100-05/07	M1900	MATH245	W1900 R1300	MF1000 W1000 CSCI308 POL5140	MWF1400 GEOL203 ECON203
3:30 PM	TR1300 T1300 MATH211	MATH216 MATH240/MATH241 MWF0830 GEOL250 WF1330	TR1000 MECH202 ENGR101 WF1200	TR1500	MW1200 MWF1200 TR0900 MWF1600	MATH222 R1900 CHEM211 W1300 ECON281
7:30 PM	MATH192 MATH202 MATH201 GEOL204	PHYS212P WF1300 ECON237		TR1100 ECON101 ECON202 WF1500	MWF0900 CSCI203 CSCI204 ENGR229 POL5120-01/02 M1000 WF1000	

Analysis Summary 4.

Drag and Drop to modify the schedule.

Reload the page to view the new version in the drop-down

Issues	Cases
Overlaps	12
Unavoidable Overlaps	11
Three exams in 24 hours	33
Four exams in 48 hours	10
Back to back exams	237

1. **Return to Dashboard:** returns you to the previous screen, the Dashboard

2. **Create Spreadsheet:** Create a .csv similar to the large PDF table that is posted for students
3. **Download:** Create a PDF with a table similar to the smaller table posted for students
4. **Analysis Summary:** This table shows what problems this schedule has.
 - Overlaps: a student is scheduled to be in two exams at the same time
 - Forced Overlaps: overlaps caused by how courses are grouped. Caused by a student being in 2 courses that are in the same course group.
 - Three Exams in 24 hours: a student has at least 3 exams within 24 hours
 - Four Exams in 48 hours: a student has at least 4 exams within 48 hours
 - Back-to-back exams: a student has two exams in a row on the same day
 - Night to Morning: A student has an exam during the last slot of the day, then an exam in the first slot of the next day. This does not count as a back-to-back exam.
 - Faculty Overlaps: A faculty member is scheduled to be in two exams during the same slot. This does not include two courses that have the same name, as they are assumed to be cross-listed courses which only have one exam
 - Faculty back-to-backs: A faculty member has an exam, and also has an exam in the next time slot. This includes both back-to-back and night-to-morning, which is different from how students are calculated.
5. **Schedule Title:** name of the schedule being viewed
6. **Semester:** which semester this schedule was made from
7. **Version name:** name of the current version of this schedule being viewed
8. **Select version:** click to open a drop-down menu to select a different version of the schedule
9. **Preference Profile:** click to show what preference profile was used in optimization
10. **Constraints:** click to show what constraints were used in optimization
11. **Schedule Table:** This is the detailed view of where the application placed specific courses in the schedule. This table is interactive, allowing you to drag courses to different places in the schedule. To see what effect these changes had on the schedule, click the reevaluate button under the Analysis Summary section.
12. **Student Conflict Details** (Not shown, at the bottom of the page): This section gives a more detailed view of the numbers provided in the Analysis Summary table. It provides the Student ID and information about the conflict that the student faces. The Important sections are student_overlap and three_in_24, which are important enough for students to request changes to their schedule.

The student IDs are **anonymous**, so they cannot be directly used to locate which student the conflict is about.

Schedule Versions

When editing a schedule, you might want to save the exact layout you are looking at. This is enabled by schedules having multiple versions. When you are looking at a schedule you want to keep a copy of, you should create a new version of it. This can be done by the following steps:

1. If not done already, click the “evaluate” button near the top right of the screen in the “Analysis Summary” section.
2. Now that the schedule is evaluated, click “Save the schedule” to create a new version of the schedule.
3. **Reload** the page
4. Now, under the “select version” drop-down, there should be a new entry titled “new version”, click this inside the drop-down to select it.
5. Rename the version by clicking the blue pencil in the row titled “Version Name”. Once clicked, type in a name you like, then hit enter to set the name.

Saving Schedule Data

Here is how to view the spreadsheet from the Create Spreadsheet button on the Analysis page, with individual course information/exam slots created by the web application tool.

1. Go to the Analysis page
2. Create Spreadsheet button (Button 2 in the screenshot below)

The screenshot shows two side-by-side panels. The left panel, titled 'Schedule Details', contains a table with three rows: 'Schedule Title' (value: 'grasp testing part a billion'), 'Semester' (value: 'Spring Term 2023-2024'), and 'Version Name' (value: 'New Version' with a blue pencil icon). Above the table are three buttons: 'Return to Dashboard' (green), 'Create Spreadsheet' (blue), and 'Download' (blue). The right panel, titled 'Analysis Summary', contains instructions: 'Drag and Drop to modify the schedule.' and 'Reload the page to view the new version in the drop-down'. Below the instructions are two tabs: 'Issues' and 'Cases'. Red numbers 1 through 7 are overlaid on the image to indicate specific steps: 1. points to the 'Return to Dashboard' button; 2. points to the 'Create Spreadsheet' button; 3. points to the 'Download' button; 4. points to the 'Analysis Summary' header; 5. points to the 'Schedule Title' row; 6. points to the 'Semester' row; 7. points to the 'Version Name' row.

Schedule Details	
Schedule Title	grasp testing part a billion
Semester	Spring Term 2023-2024
Version Name	New Version

Return to Dashboard

Create Spreadsheet

Download

Analysis Summary

Drag and Drop to modify the schedule.

Reload the page to view the new version in the drop-down

IssuesCases

3. The downloaded file is a CSV file. We recommend using Microsoft Excel for viewing this file. If you are running this web application tool on an external remote desktop server, Microsoft Excel may not have been installed on the server. In this scenario, you may send the file to your main laptop or computer via email or shared folder so that you can open the downloaded file using Microsoft Excel.

Example Workflows

1. You want to explore what would happen if each day had 5 exam times.
 - a. Create a new semester using the same information as normal, but put 5 exam times into the “exam times” field during the creation.
 - b. Create a schedule with this new semester to see how things play out
2. You notice that students start caring more about particular issues with exam times. For example, students start to really hate night to morning.
 - a. Create a new preference profile (found in the “Manage Preference Profile” button on the schedule creation page)
 - b. For this new preference profile, copy and paste the values that you have been using. These can be taken from a different preference profile.
 - c. **Increase** the value for the problem in question. For this example, if night to morning was originally 0.064, increase its value to be 0.07. The higher the value, the less the optimizer will take them.
 - d. Create and optimize a new schedule using the preference profile created
3. You have a schedule that you really like, but a new constraint comes up later
 - a. You have two options: either use the drag and drop to change the schedule or create a new schedule with the new constraint added.